



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Faculty of Engineering, Built Environment and Information Technology

Fakulteit Ingenieurswese, Bou-omgewing en
Inligtingtegnologie / Lefapha la Boetšenere,
Tikologo ya Kago le Theknolotši ya Tshedimošo

Unit 9: Project Network Diagram and Related Metrics

BSS 310 Engineering Management 2026

Lecturer: Prof Michael Ayomoh



Quick Recap/Questions arising from Previous Lecture

The lecture addressed:

- the concept of project phases/deliverables
- the fact that a scope statement forms the basis for setting up project phases and deliverables
- Work Breakdown Structure (WBS) and overall project planning
- the principles and purpose of a Work Breakdown Structure (WBS) and how to develop a Work Breakdown Structure (WBS)
- how to develop a Project Responsibility Assignment Matrix
- how these whole concepts fit into the semester project

Learning Outcomes for the Current Lecture

The expected learning outcomes at the end of this lecture include:

- Demonstration of understanding and ability to construct and analyse a network diagram from a WBS
- Ability to analyse and evaluate the early start and finish dates using the forward pass; the late start and finish dates using the backward pass
- Ability to identify the critical path in a network diagram and calculate the total float and free float from network activities

Presentation Outline

The expected learning outcomes at the end of this lecture include understanding how to:

- draw a network diagram from a WBS
- calculate early start and finish dates using the forward pass
- calculate late start and finish dates using the backward pass
- identify the critical path in a network diagram and
- calculate the total float and free float of activities

How can project scheduling be done?

1. By guessing
2. By knowing
3. By allowing the project to take the time it needs
4. By calculating

Project
scope



Project
phases and
deliverables



Work
breakdown
structure



Activity
sequencing



Network
diagram

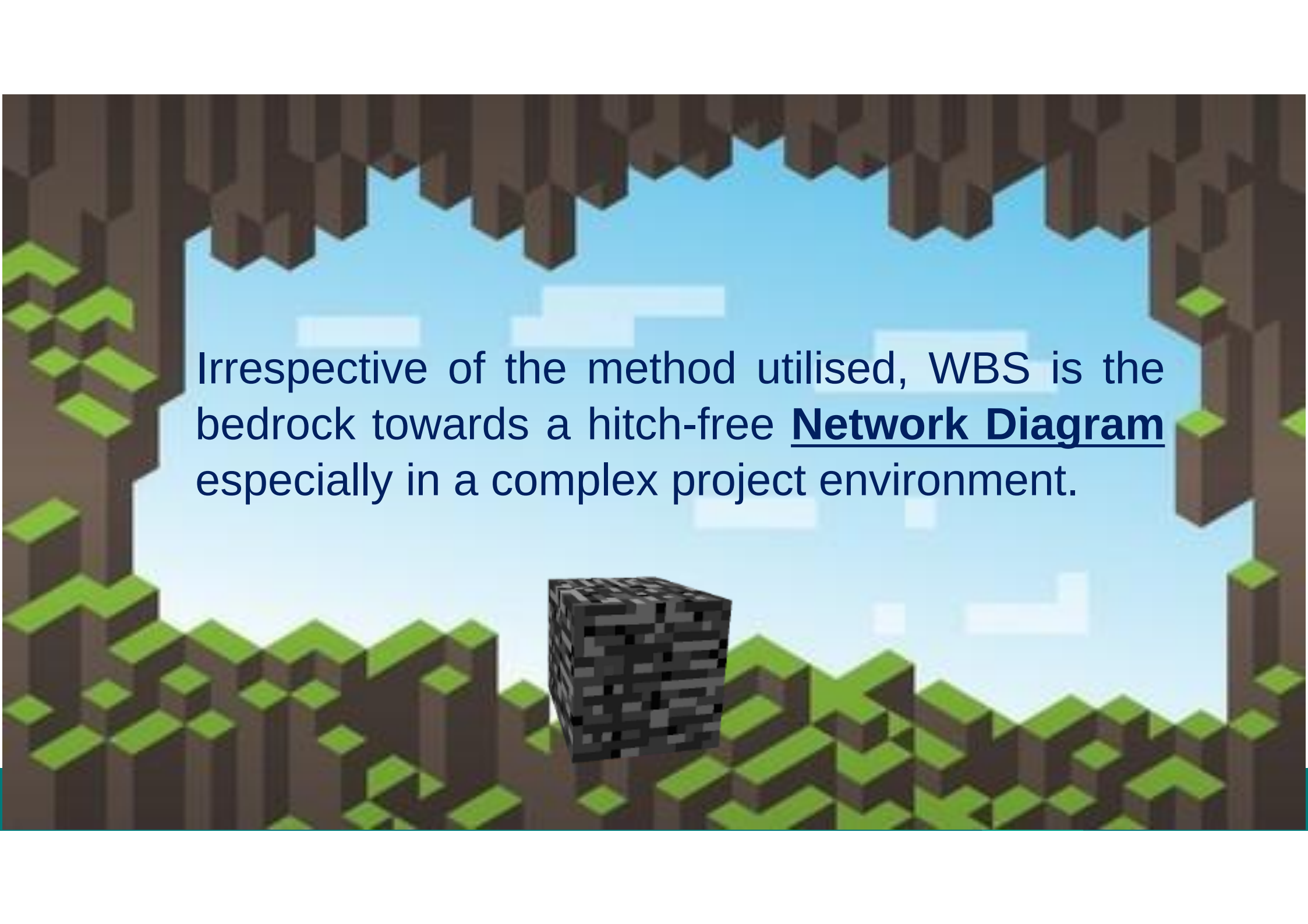


Project
budget



What happens if time Planning is not Accurate or Poorly Done?

- Resources wait (bottlenecks)
- Project delays
- Financial implications (profit, penalty fees, cost of resources)
- Unhappy people (Increased stress)
- Project will not deliver what was promised

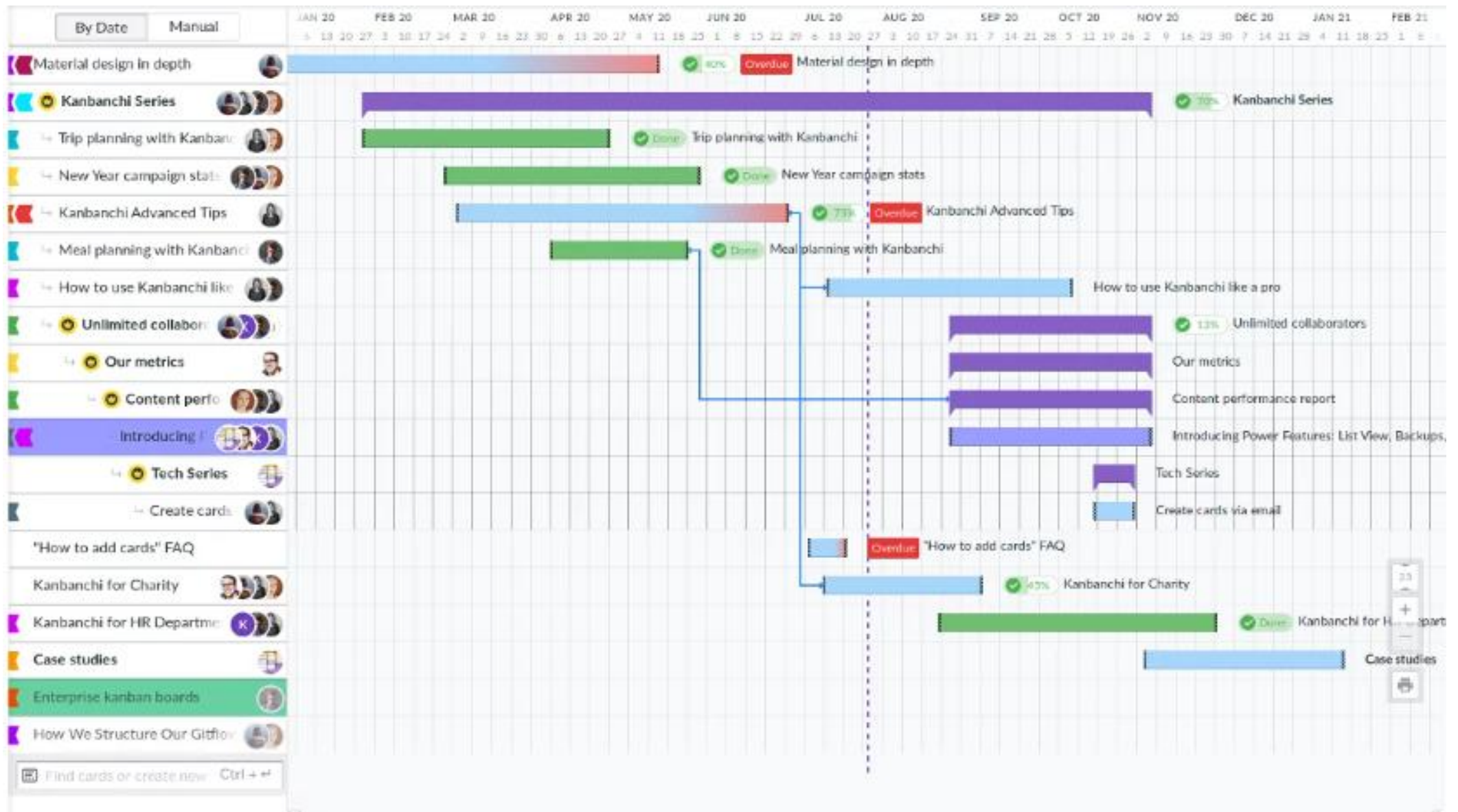
The background is a pixelated, isometric landscape from the game Minecraft. It shows a cave entrance with dark brown dirt and green mossy dirt blocks. A single cobblestone block is placed on the ground in the foreground. The sky is light blue with white pixelated clouds.

Irrespective of the method utilised, WBS is the bedrock towards a hitch-free **Network Diagram** especially in a complex project environment.

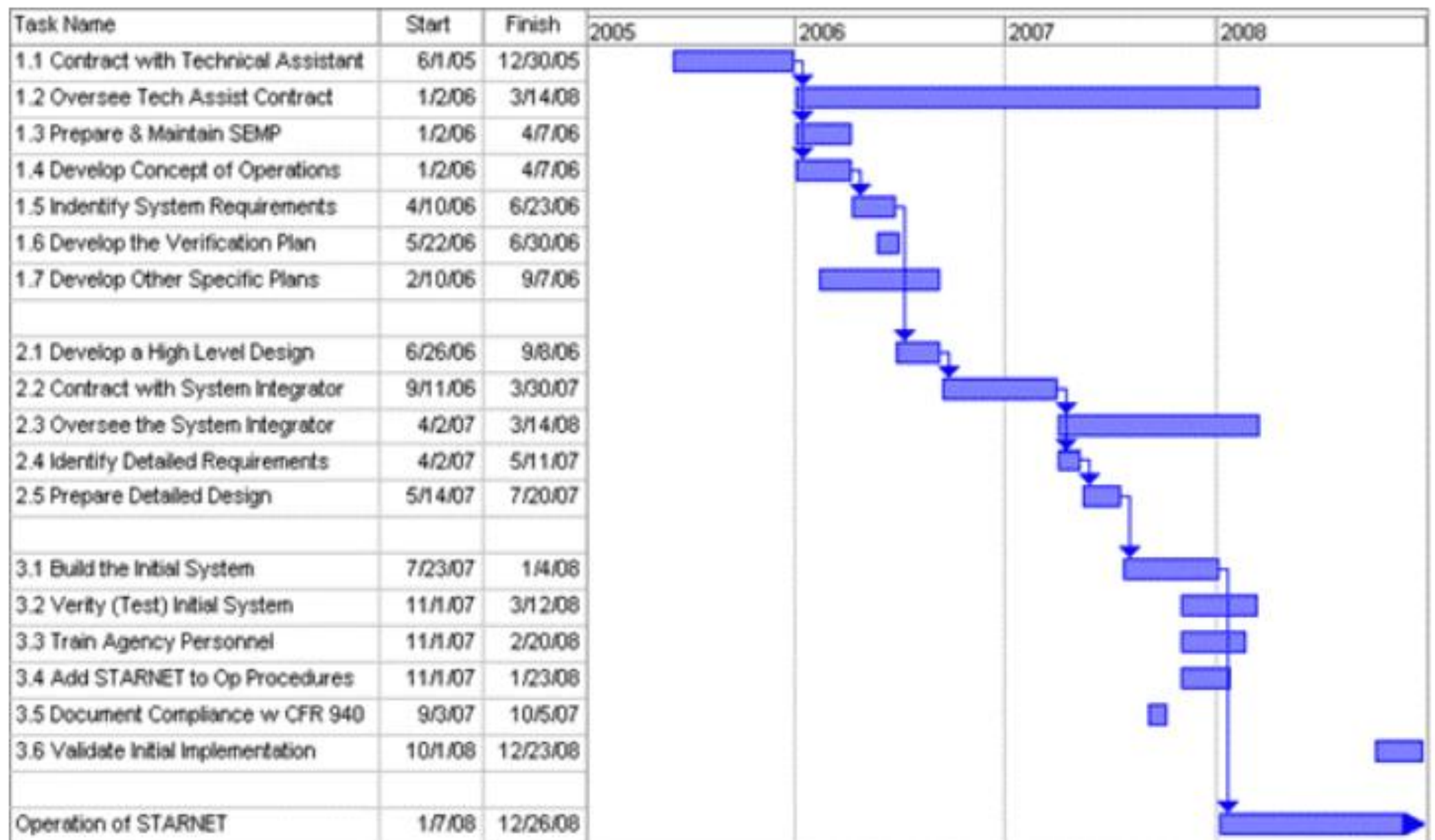
Gantt Chart

- A Gantt chart is **a *project scheduling tool***, ideal for a relatively less complex, short-term projects presenting the timeline of events
- A Gantt chart doesn't reflect the best usage of resources or the order of tasks to completion (in terms of critical or shortest path)

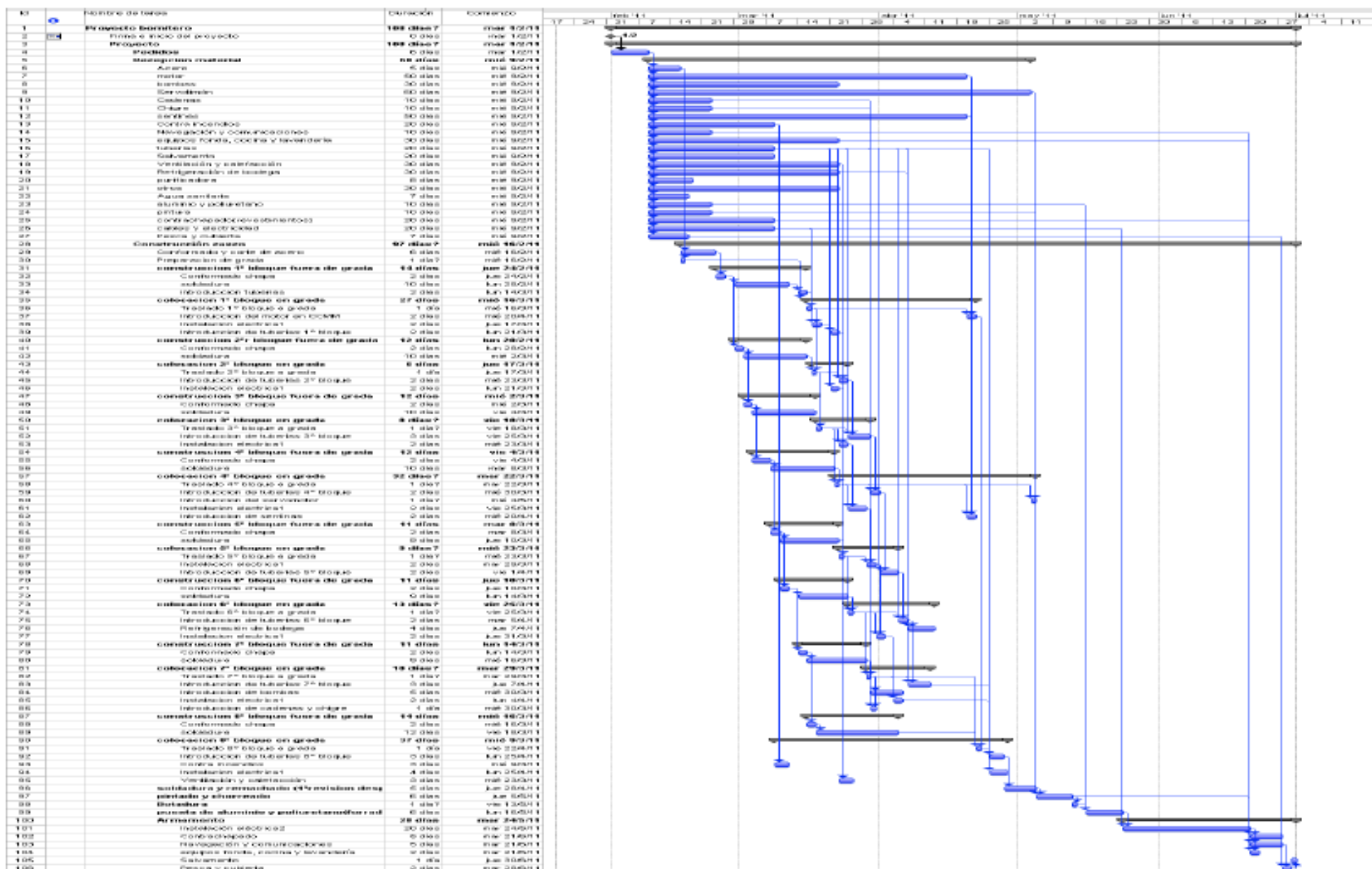
		DAYS																																							
Activity	Responsible	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
		M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F	S	S	M	T	W	T	F
Activity 1																																									
Activity 2																																									
Activity 3																																									
Activity 4																																									
Activity 7																																									
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Activity 9																																									
Activity 10																																									
Activity 5																																									
Activity 6																																									



Gantt Chart



Gantt Chart



Precedence Network Diagram

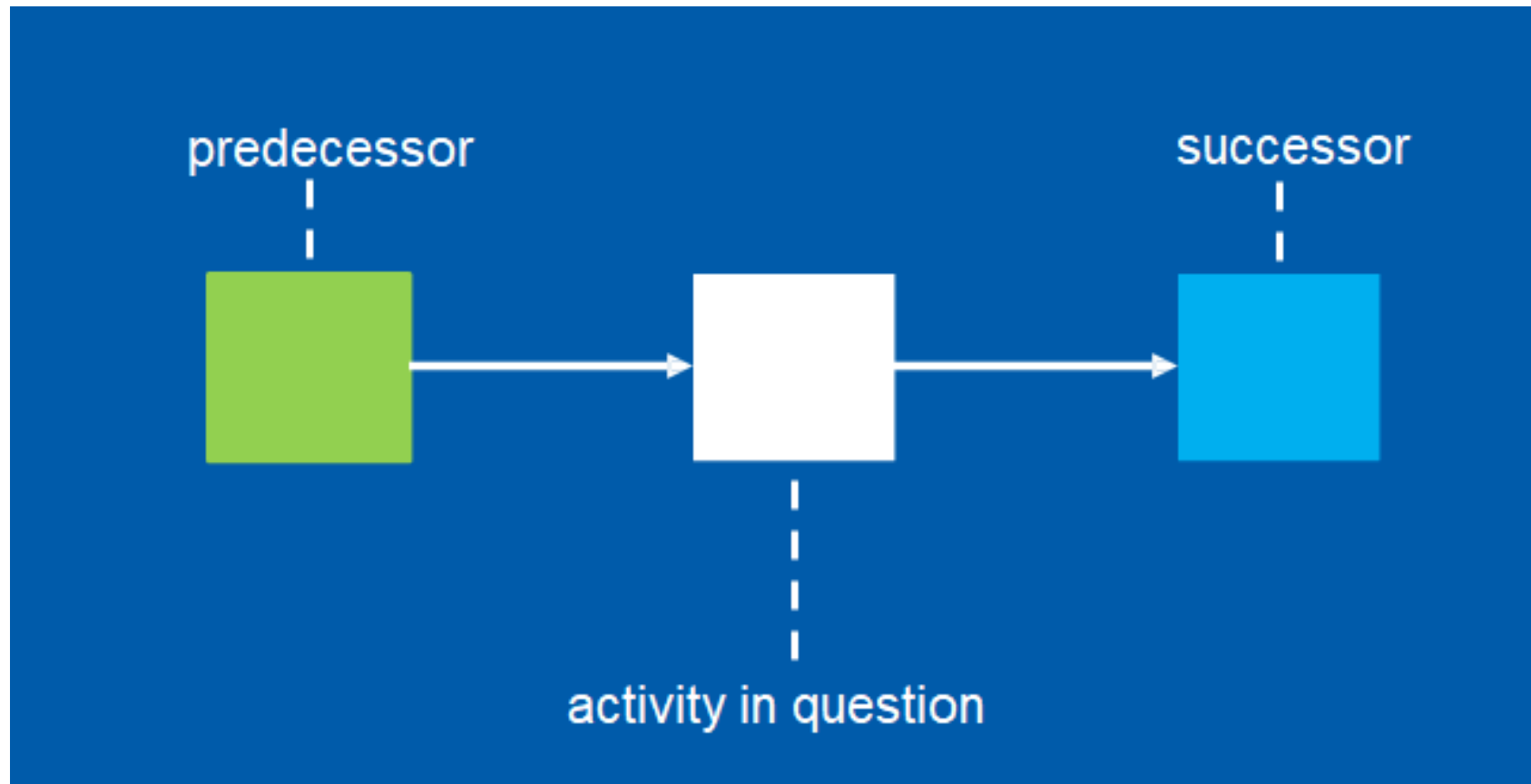
- **Critical Path Method**-This is a project activity scheduling technique that assigns and utilizes a single duration per activity and in addition, identifies a critical path in the network diagram i.e. a path made up of activities that cannot be delayed. It is often used when the durations are certain.

Precedence Network Diagram

- **Project Evaluation and Review Technique (PERT)**-This is a project activities scheduling technique that assigns and utilizes the average of three estimated durations per activity i.e. the worst, most probable and best times.

In addition, it identifies a critical path in the network i.e. a path made up of activities that cannot be delayed. It is often used when the durations are not certain or tasks are somewhat improperly defined.

Some Notation Rules

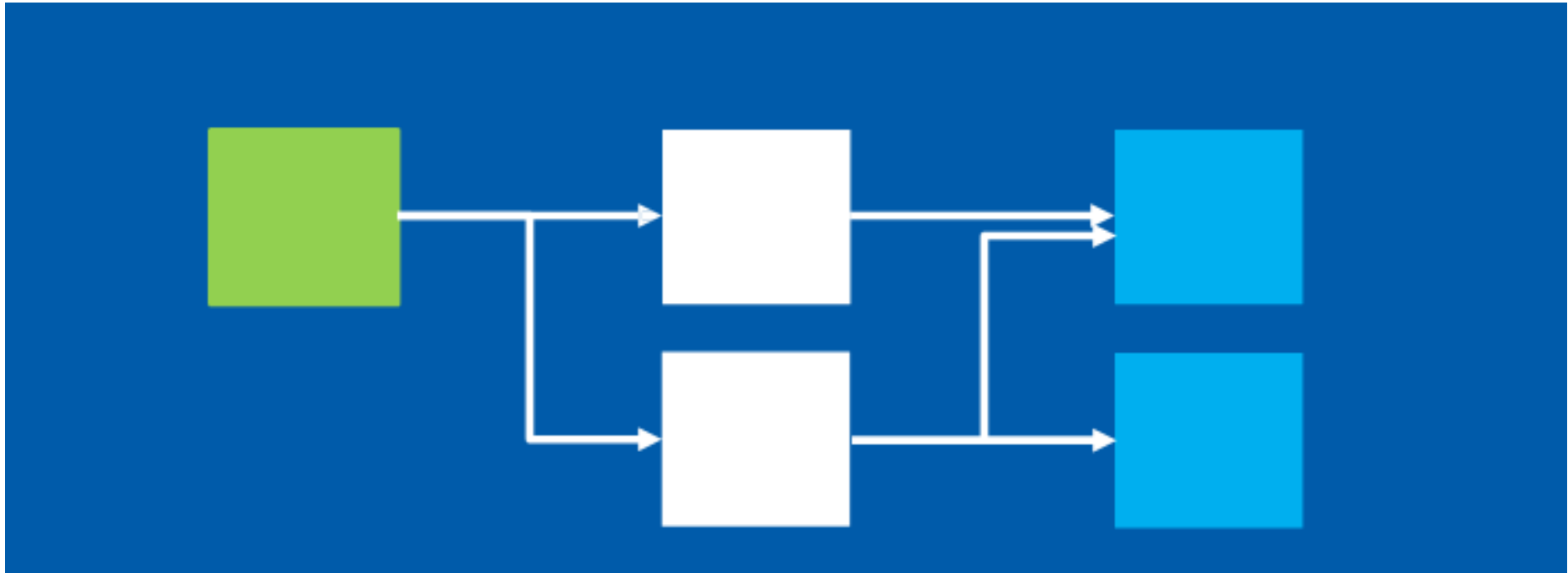


Some Notation Rules: arrow entry and exit



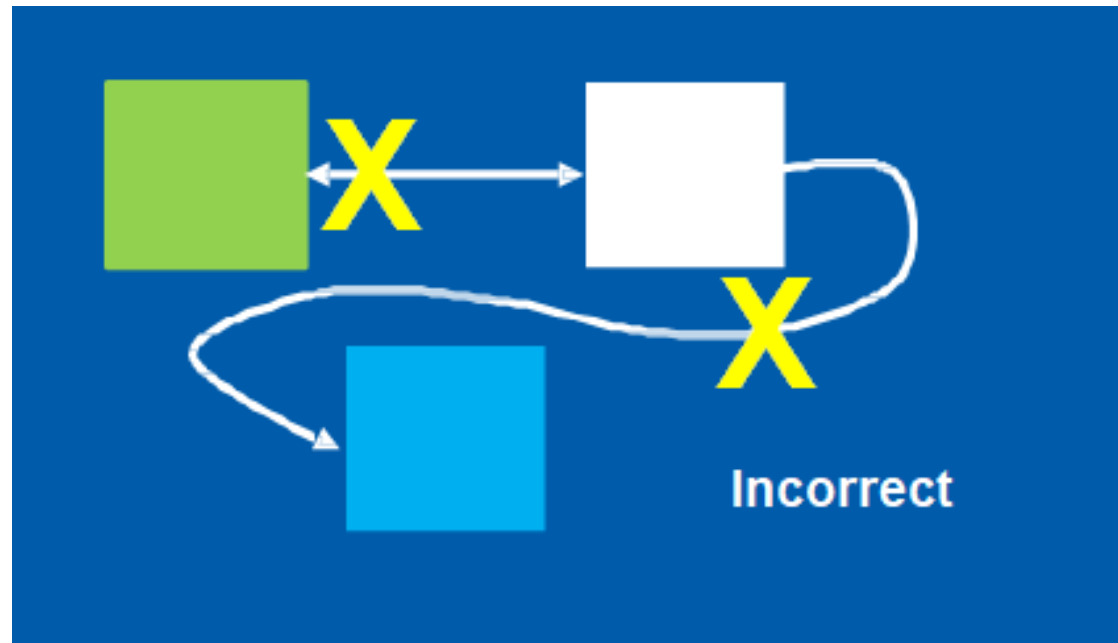
Correct: Enter from left, exit to right

Some Notation Rules: direction of flow

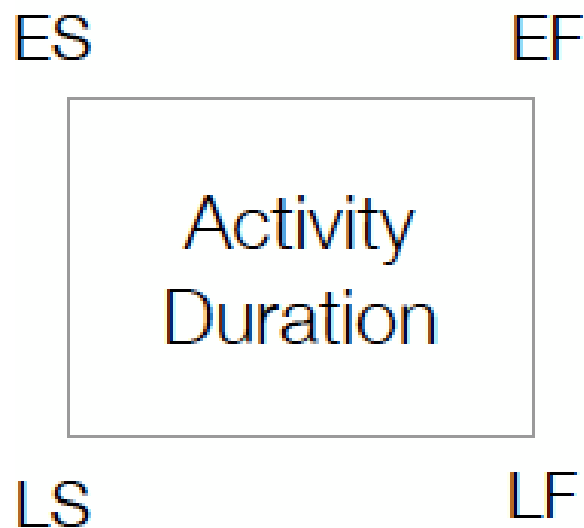


Correct flow: left to right

Some Notation Rules: direction of flow



The Basic Concept of Project Time Planning: Precedence Diagramming Model (PDM) Approach



ES = Earliest Start

EF = Earliest Finish

LS = Latest Start

LF = Latest Finish

Network diagram (forward and backward pass)

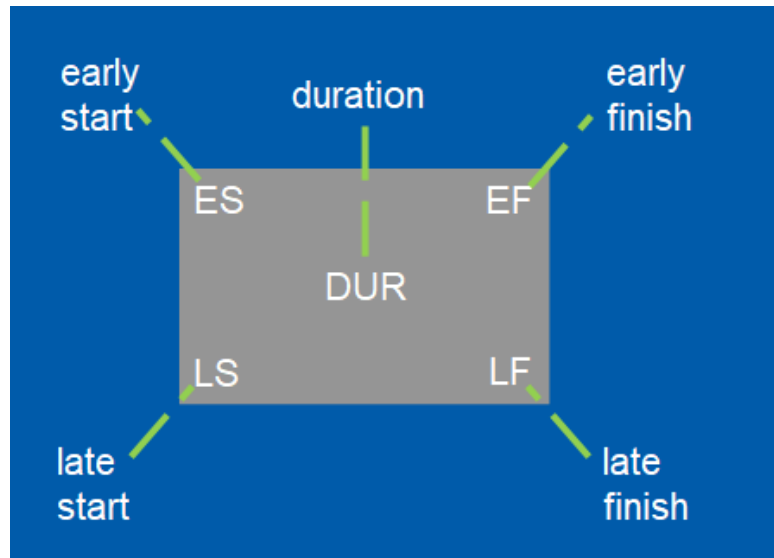
Forward pass

- Earliest project finish date
- Earliest start and finish dates for activities
- Work forward through network starting TODAY
- Projects to be completed ASAP

Backward pass

- Latest project start date
- Late start and finish dates for activities
- Work backwards through a network from an end date
- Projects with a due date

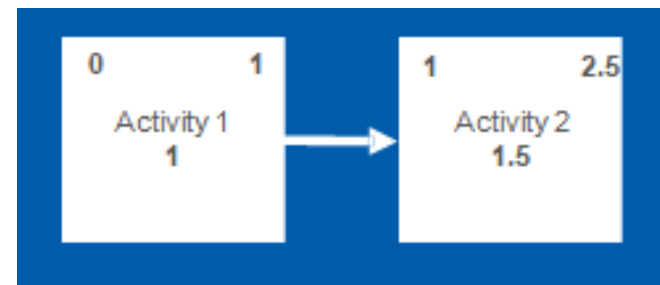
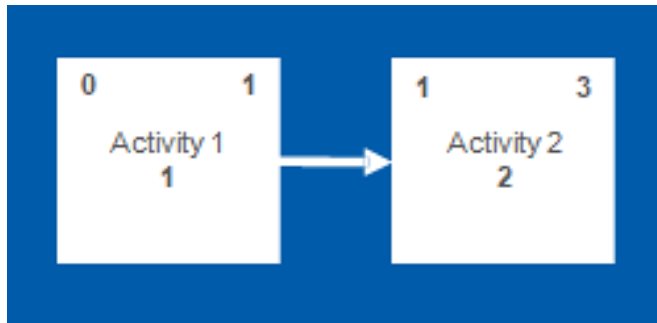
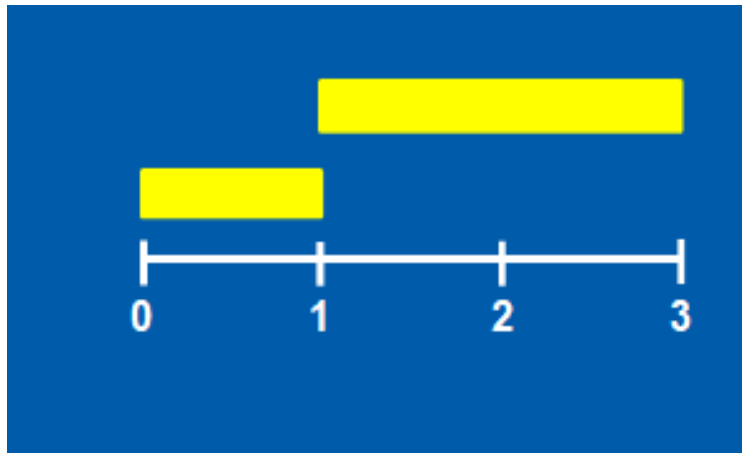
Precedence/Network diagram (forward and backward pass)



Forward pass: **early finish** = early start + duration

Backward pass: **late start** = late finish - duration

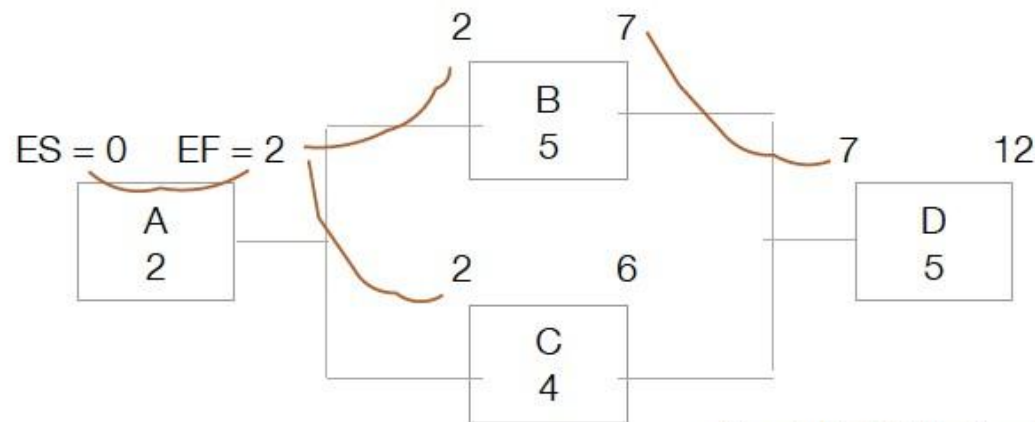
Network diagram (forward pass)



Precedence/Network diagram (forward pass)

Using a project with four activities as an example how to calculate the schedule by using the Activity-on-node Diagram. Step 1: Calculate the Earliest Start and Earliest Finish for each activity, from the first to last.

Activity	Duration [Days]	Preceding Activities
A	2	
B	5	A
C	4	A
D	5	B,C

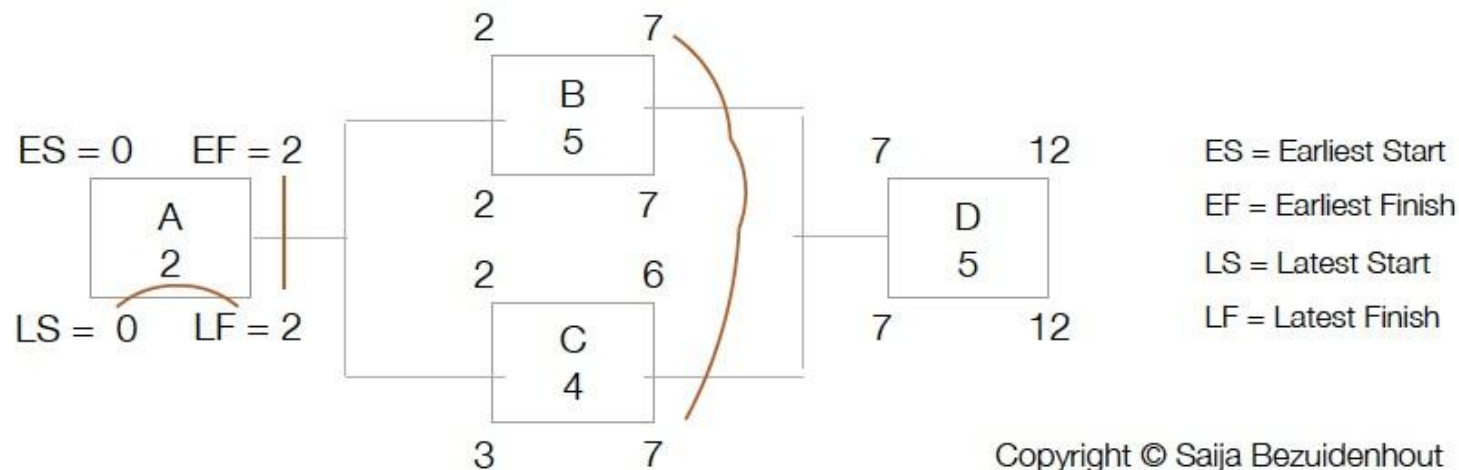


ES = Earliest Start
EF = Earliest Finish
LS = Latest Start
LF = Latest Finish

Precedence/Network diagram (backward pass)

Step 2: Calculate the Latest Start and Latest Finish for each activity from the last to first activity.

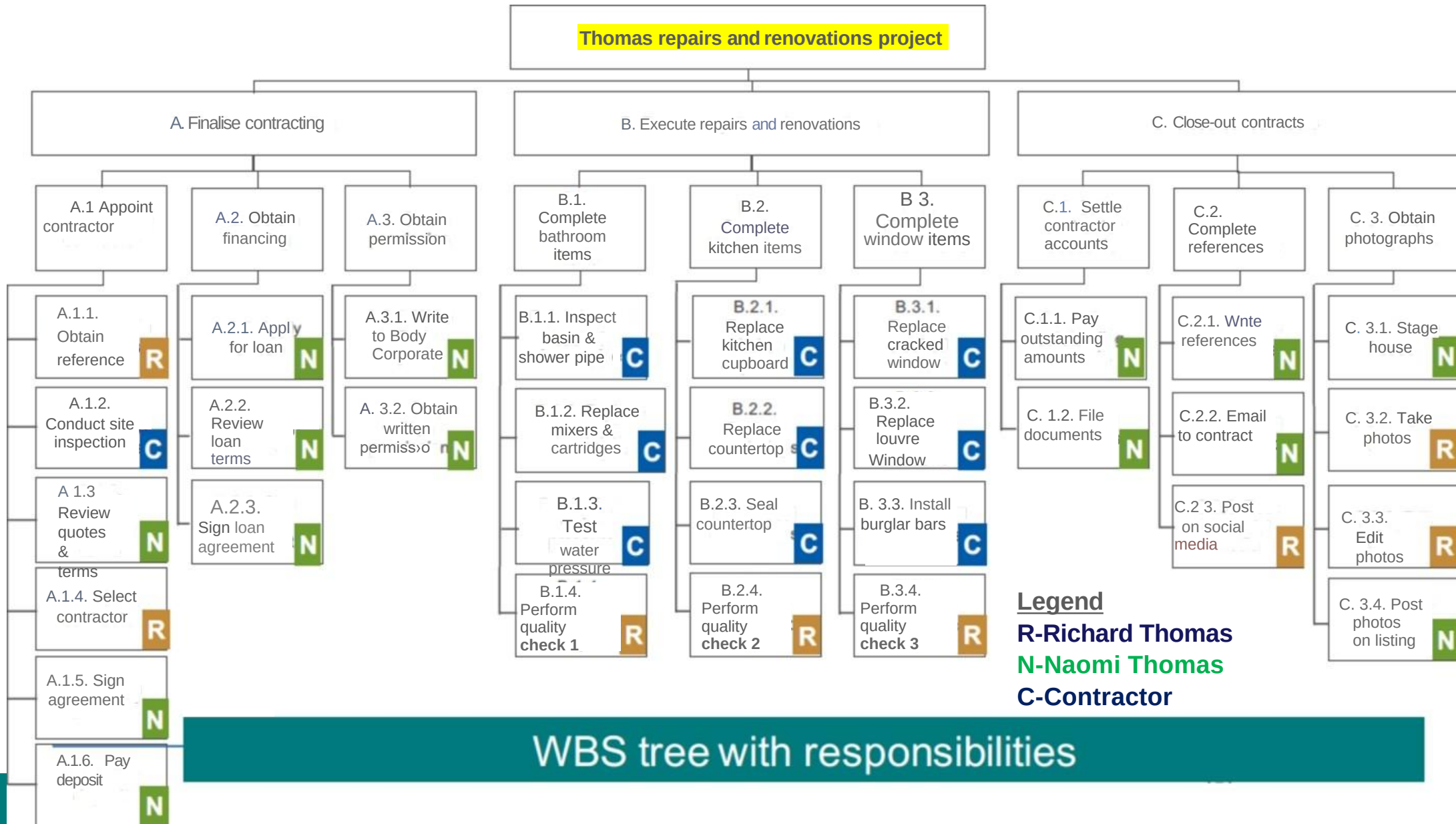
Activity	Duration [Days]	Preceding Activities
A	2	
B	5	A
C	4	A
D	5	B,C



Network Diagram Construction (Based on WBS from the previous lecture)

Phase	Deliverable	Due date
Contracting	Body corporate permission	1 December 2019
	Agreed quotations and service agreements	20 December 2019
	R40 000 available to project	12 December 2019
	Deposit paid to contractor(s)	27 December 2019
Execution	All repair work completed	28 February 2020
	All renovation work completed	28 February 2020
	All quality assessments passed	28 February 2020
Close-out	All contractor accounts settled	6 March 2020
	Written references of work	6 March 2020
	Professional photographs	28 February 2020

Work Breakdown Tree with Responsibilities

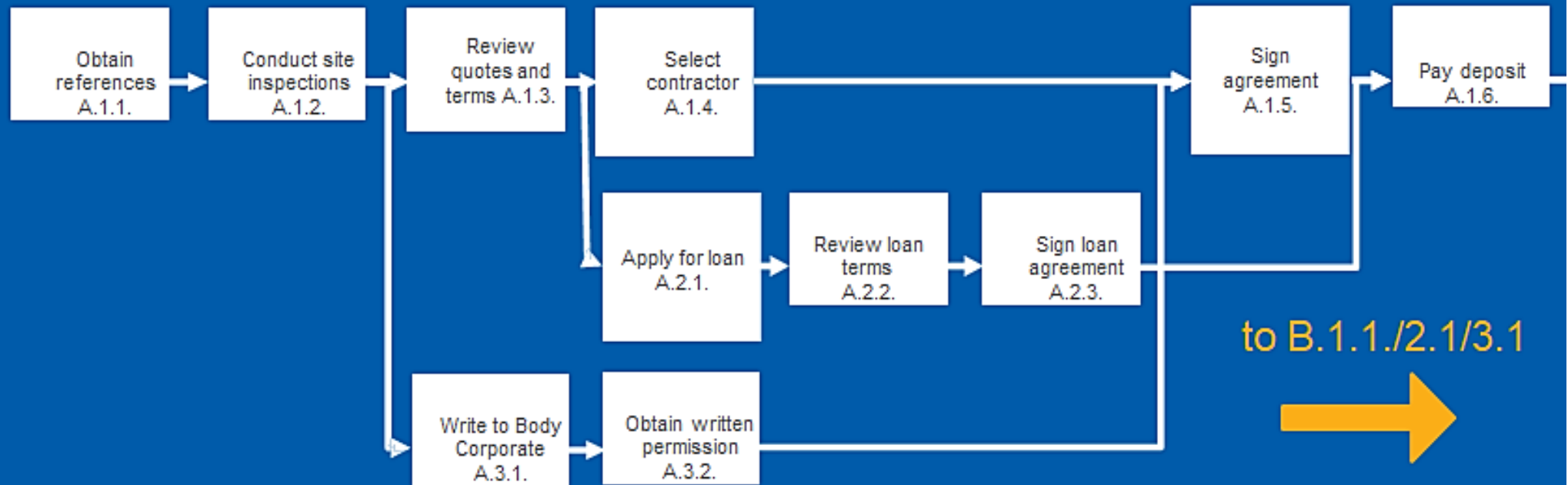


Activity (from WBS)	WBS #	Predecessors	Duration (days)
Obtain references	A.1.1.	none	1
Conduct site inspections	A.1.2.	A.1.1.	3
Review quotes & terms	A.1.3.	A.1.2.	2
Select contractor	A.1.4	A.1.3.	0.5
Sign agreement	A.1.5.	A.1.4., A.3.2.	0.5
Pay deposit	A.1.6.	A.1.5., A.2.3.	0.5
Apply for loan	A.2.1.	A.1.3.	4
Review loan terms	A.2.2.	A.2.1.	1
Sign loan agreement	A.2.3.	A.2.2.	3
Write to Body Corporate	A.3.1.	A.1.2.	0.5
Obtain written permission	A.3.2.	A.3.1.	2

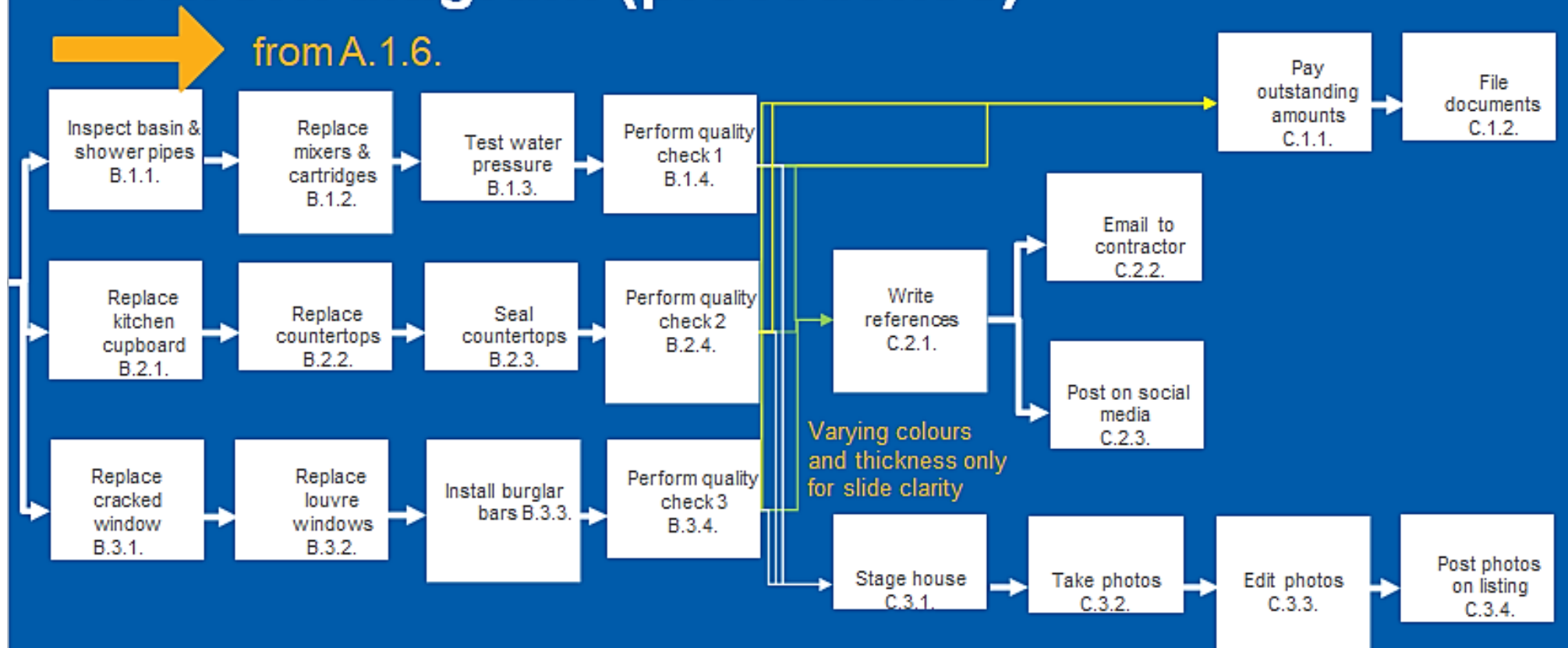
Activity (from WBS)	WBS #	Predecessors	Duration (days)
Inspect basin & shower pipes	B.1.1.	A.1.6.	0.5
Replace mixers & cartridges	B.1.2.	B.1.1.	0.5
Test water pressure	B.1.3.	B.1.2.	0.5
Perform quality check 1	B.1.4.	B.1.3.	0.5
Replace kitchen cupboard	B.2.1.	A.1.6.	4
Replace countertops	B.2.2.	B.2.1.	2
Seal countertops	B.2.3.	B.2.2.	1
Perform quality check 2	B.2.4.	B.2.3.	0.5
Replace cracked window	B.3.1.	A.1.6.	1
Replace louvre windows	B.3.2.	B.3.1	3
Install burglar bars	B.3.3.	B.3.2.	2

Activity (from WBS)	WBS #	Predecessors	Duration (days)
Perform quality check 3	B.3.4.	B.3.1., B.3.3.	0.5
Pay outstanding amounts	C.1.1.	B.1.4., B.2.4., B.3.4	0.5
File documents	C.1.2.	C.1.1.	0.5
Write references	C.2.1.	B.1.4., B.2.4., B.3.4	0.5
Email to contractor	C.2.2.	C.2.1.	0.5
Post on social media	C.2.3.	C.2.1.	0.5
Stage house	C.3.1.	B.1.4., B.2.4., B.3.4	1
Take photos	C.3.2.	C.3.1.	0.5
Edit photos	C.3.3.	C.3.2.	0.5
Post photos on listing	C.3.4.	C.3.3.	0.5

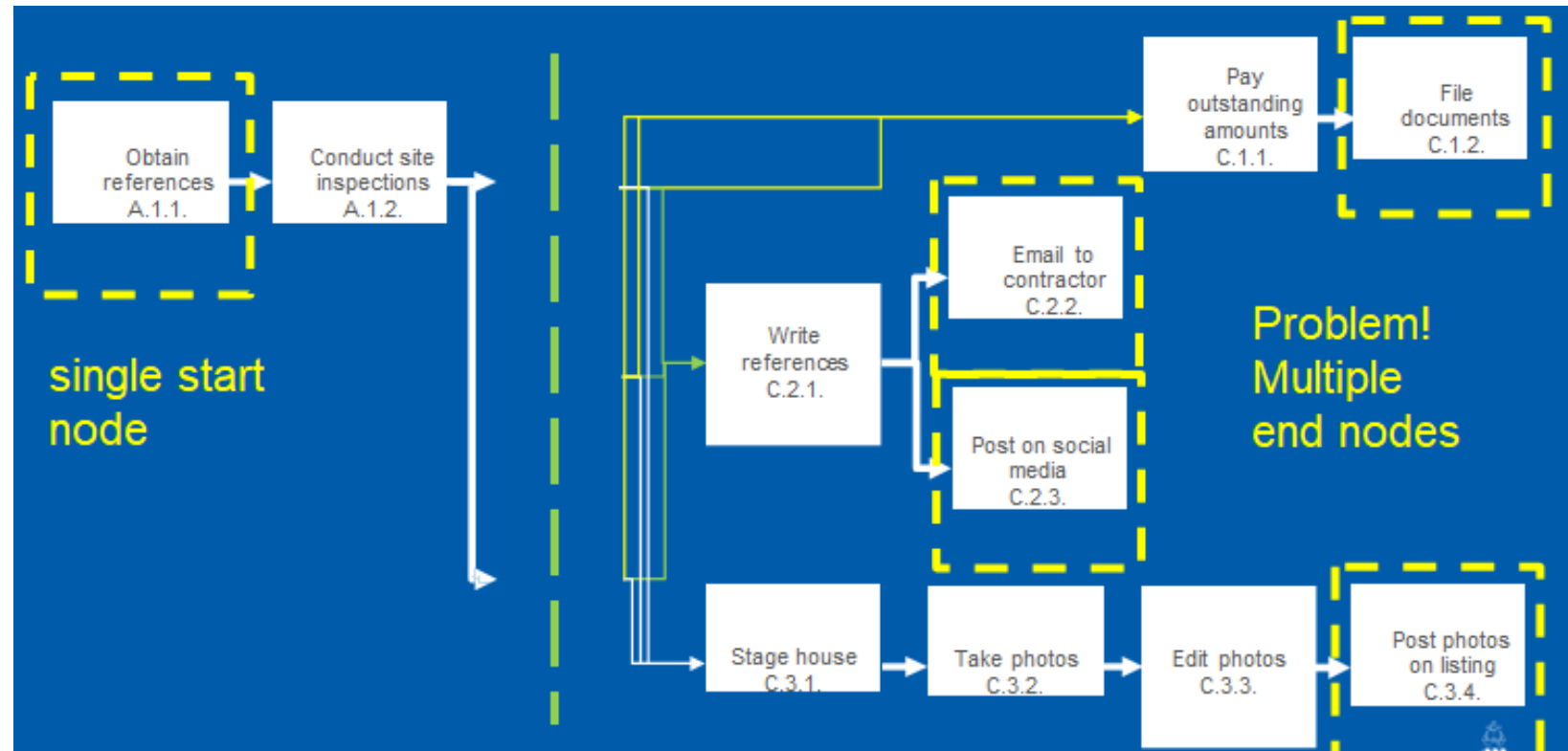
Network diagram (precedence)



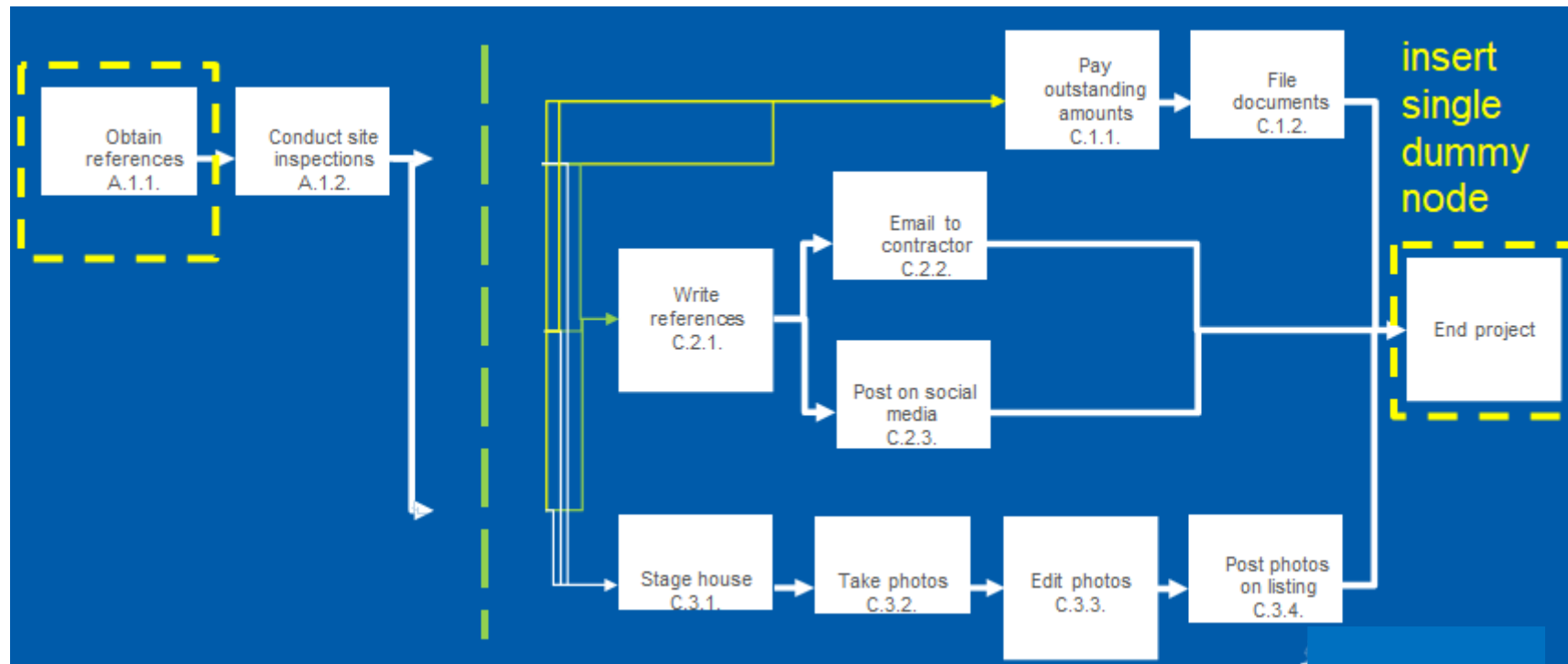
Network diagram (precedence)



Some Notation Rules: **single start and end node**



Some Notation Rules: single start and end node

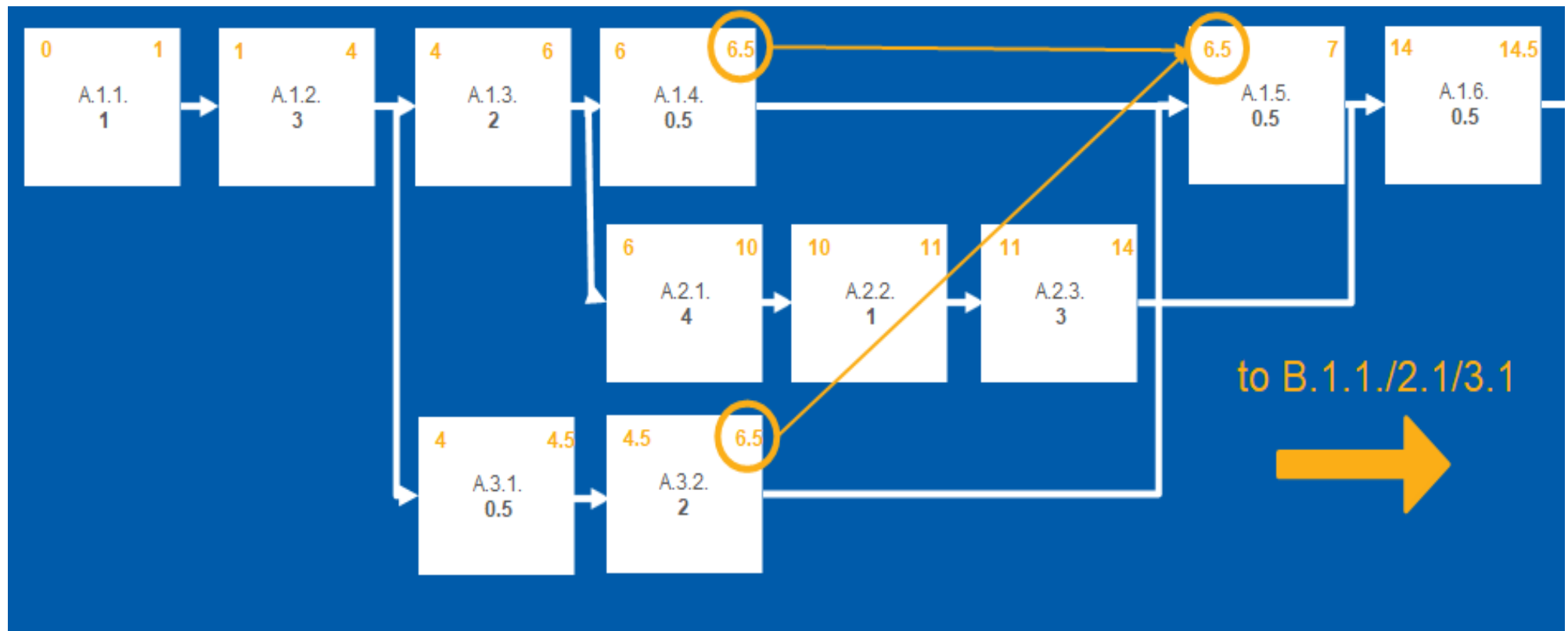


Precedence Network diagram (forward pass)



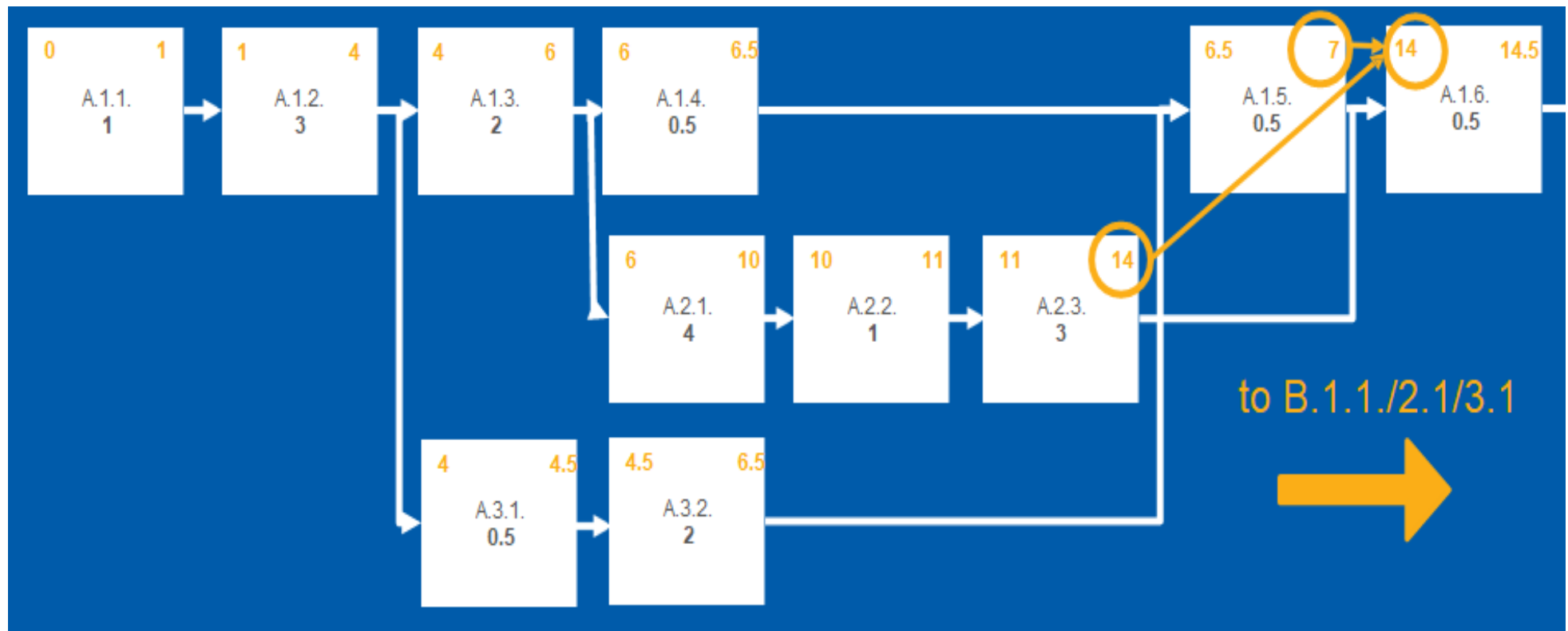
Precedence/Network diagram (forward pass)

if there are multiple predecessors, use the latest early finish as the early start.

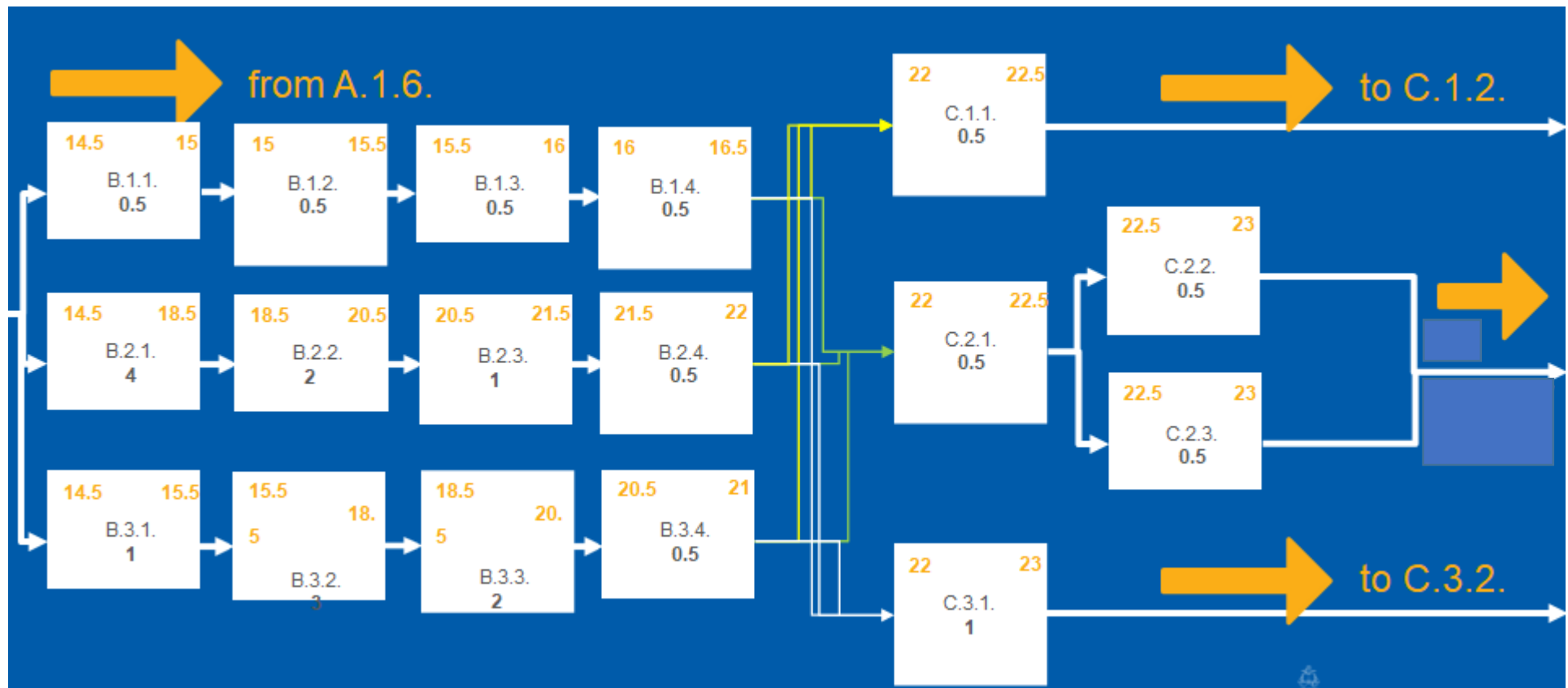


Precedence/Network diagram (forward pass)

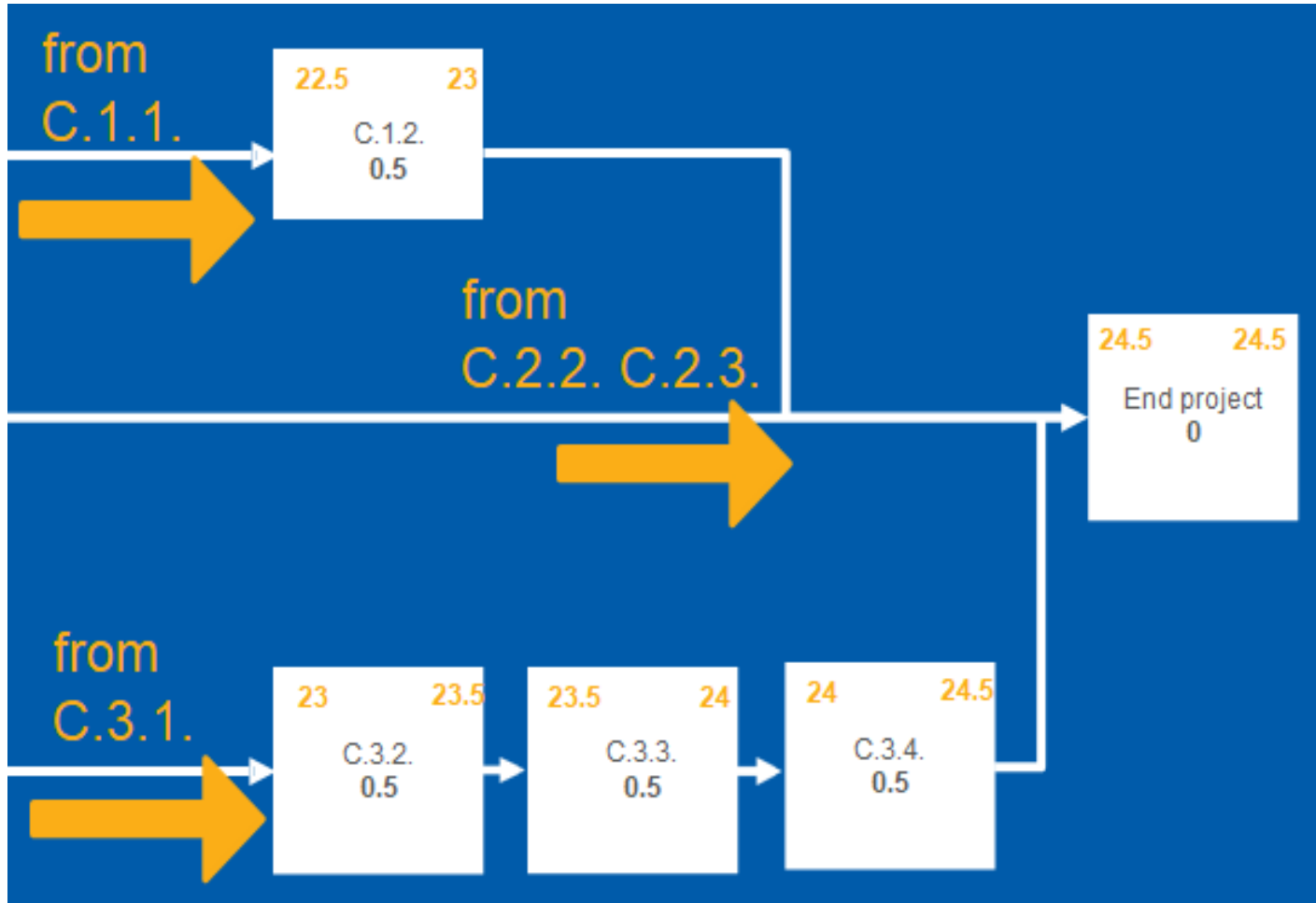
if there are multiple predecessors, use the latest early finish as the early start.



Precedence/Network diagram (forward pass)

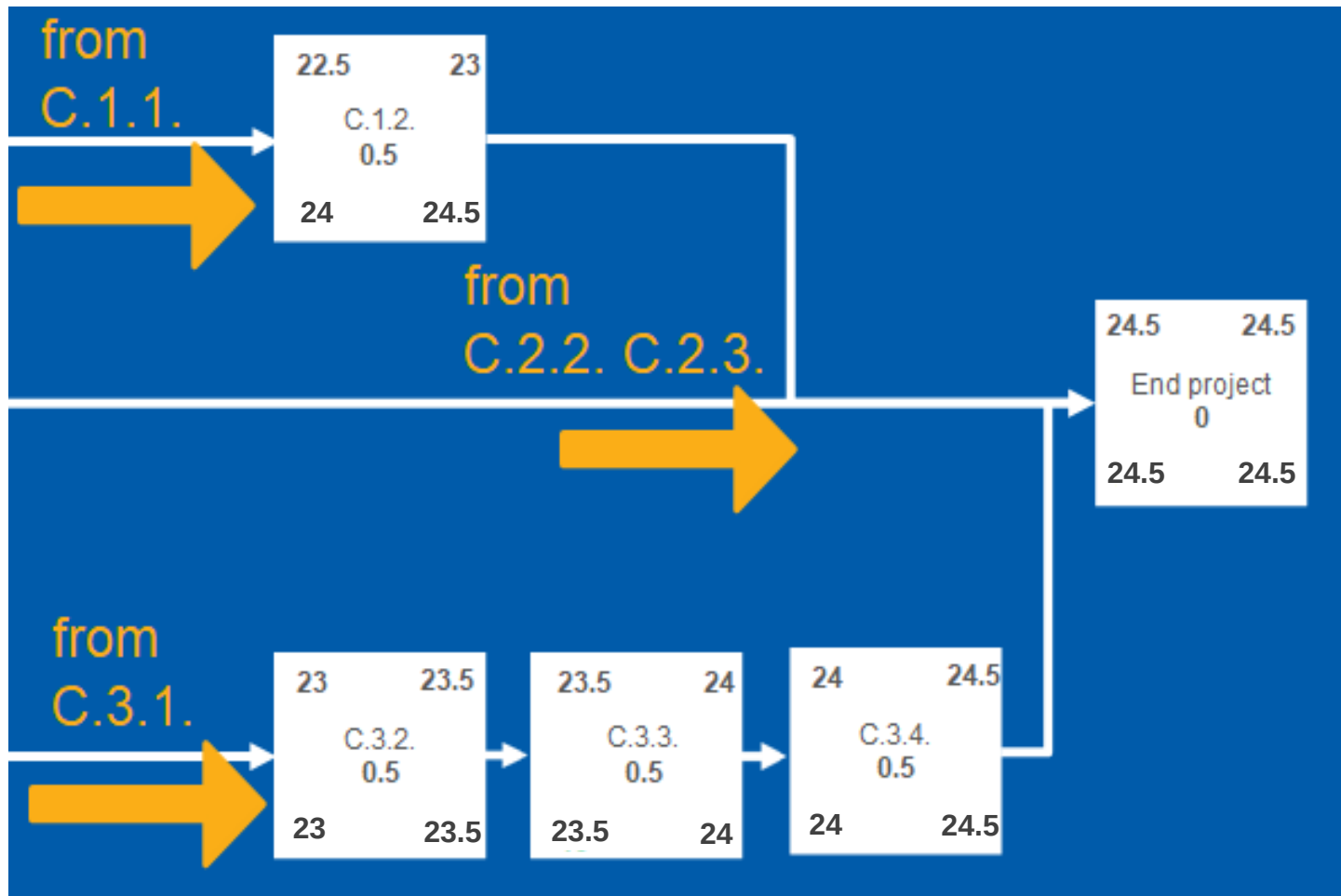


Precedence/Network diagram (forward pass)

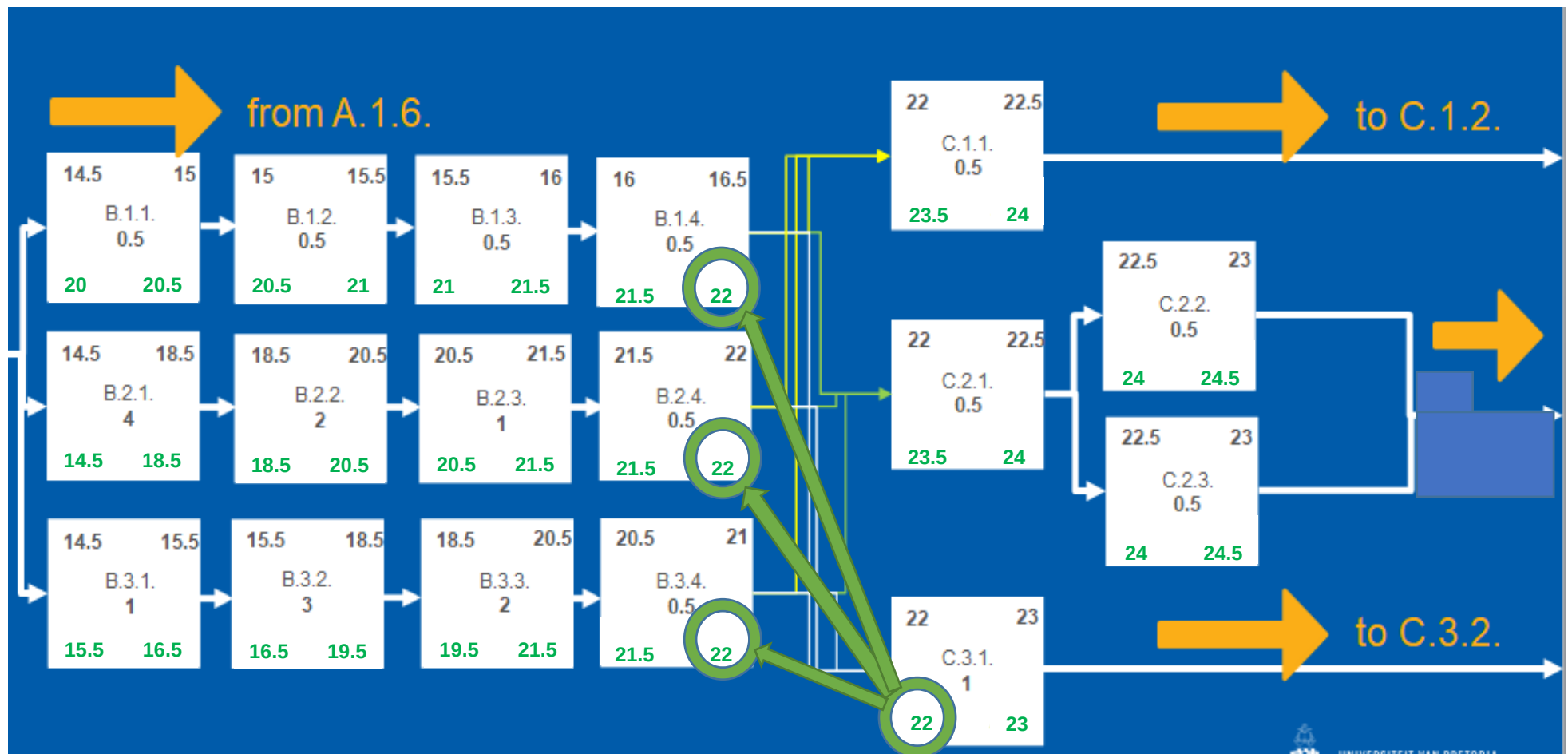


If we start the project today, the earliest we'll be done is in 24.5 days.

Precedence/Network diagram (backward pass)

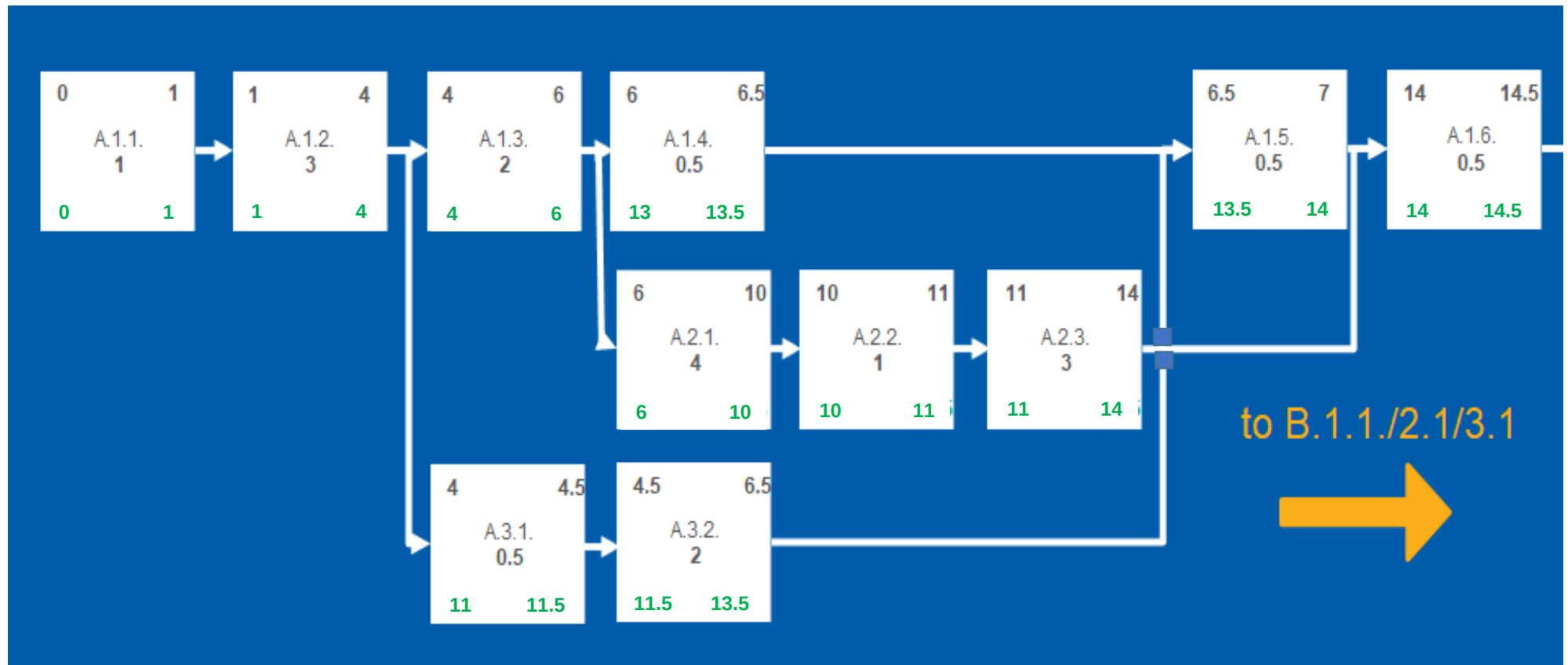


Precedence/Network diagram (backward pass)



If there are multiple successors, use the earliest late start as the late finish.

Precedence/Network diagram (backward pass)



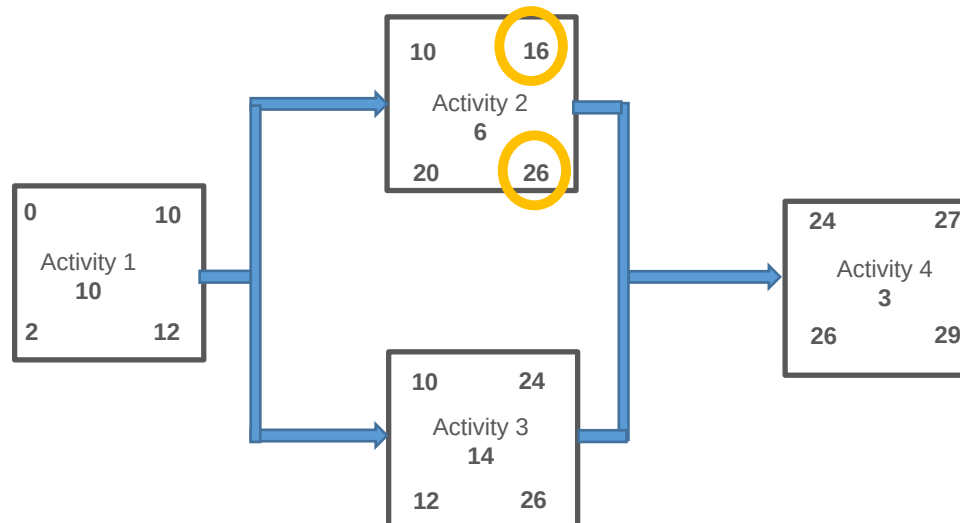
Total float

Total float: The amount of time an activity can be delayed without delaying the project finish date

$$\text{Total float} = \text{LF} - \text{EF}$$

Activity 2

$$\text{Total float} = 26 - 16 = 10 \text{ days}$$



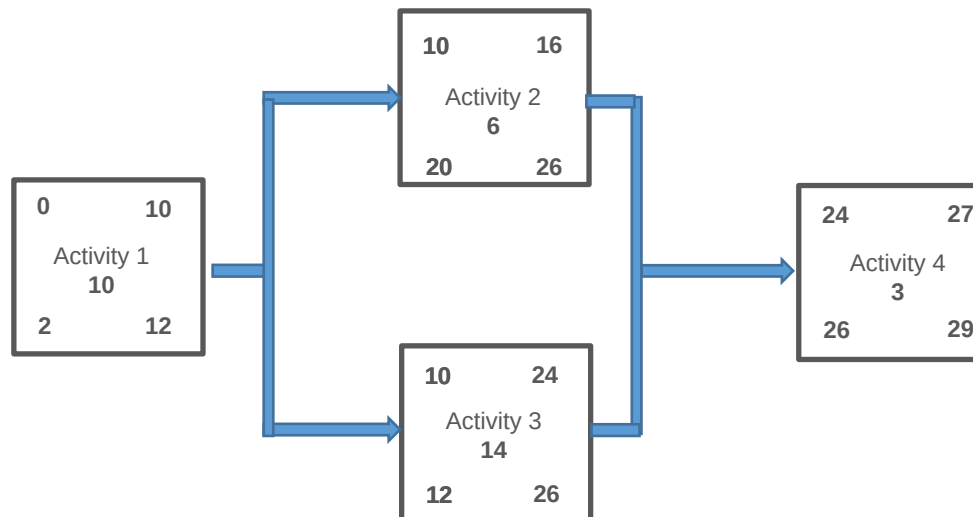
Free float

Free float: The amount of time an activity can be delayed without delaying the early start of any succeeding activities

$$\text{Free float} = \text{ES}_{\text{succ}} - \text{EF}$$

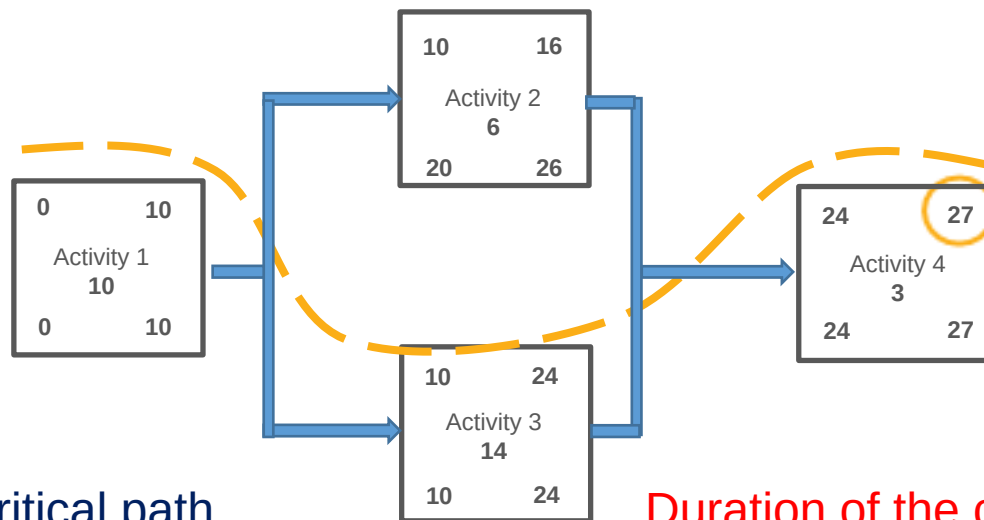
Activity 2

$$\text{Total float} = 24 - 16 = 8 \text{ days}$$



Critical Path

The critical path of a project is the sequence of activities with the longest duration. The duration of the critical path is the shortest possible time in which the project can be completed.



To find the critical path, compare the forward pass path.

Duration of the critical path = EF of last activity = 27 days.

The float for the critical path i.e. the longest path is always 0, for each activity on it.

Critical Path Identification for the WBS on slide 27

Question: What is the critical path for the network diagram constructed (slides 35 – 42) from the WBS on slide 27?

Additional Precedence Diagram Samples

Assuming table 1 represents the activity chart of a recently completed project at the Hatfield campus of TUKS, with nine (9) activities and a final completion time of 20 weeks, you are required to:

(a) Construct the network diagram of these activities.

Activity diagram

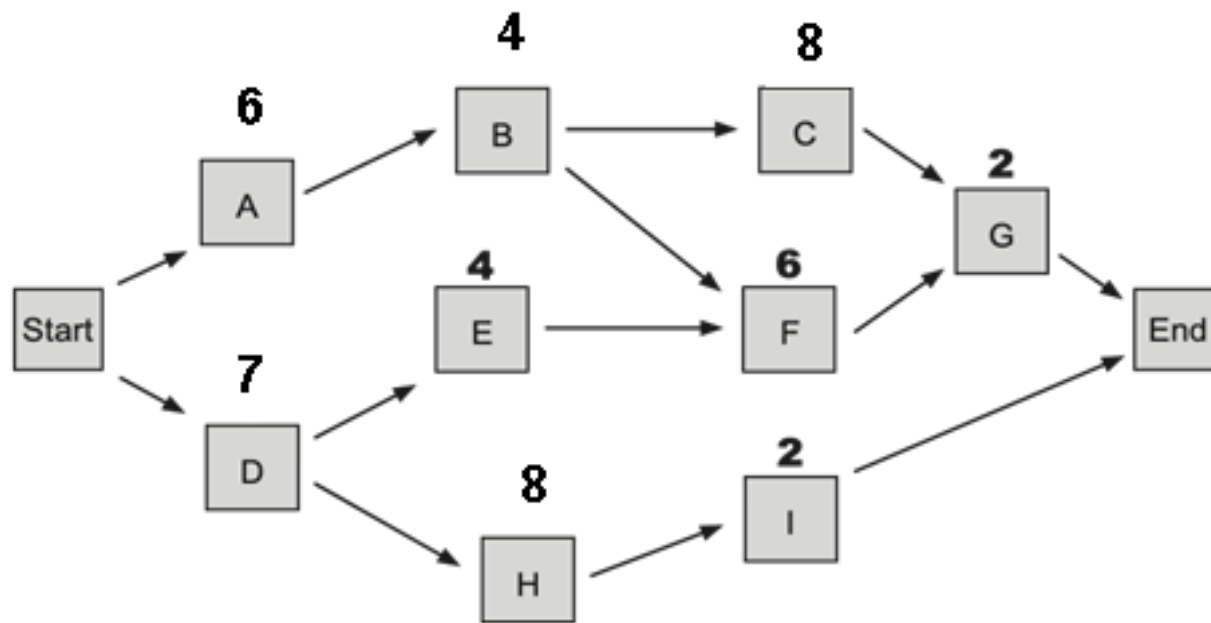
Activity	Immediate Predecessor	Duration (weeks)
A	-	6
B	A	4
C	B	8
D	-	7
E	D	4
F	E,B	6
G	C,F	2
H	D	8
I	H	2

(b). With respect to table 1, you are required to address questions (i) – (vii)

- What is the float for each activity on the critical path?
- What is the total duration for path $A \rightarrow B \rightarrow C \rightarrow G$?
- What is the total duration for path $A \rightarrow B \rightarrow F \rightarrow G$?
- What is the total duration for path $D \rightarrow E \rightarrow F \rightarrow G$?
- What is the total duration for path $D \rightarrow H \rightarrow I$?
- Which path is the critical path for the network diagram?
- Calculate the float for each activity.

Additional Precedence Diagram Samples

Solution (1a)



Additional Precedence Diagram Samples

Solution (1b)

b(i) = 0

The float is the time that it can slip before it causes the project to be delayed. A critical path is a string of activities that is longer than any other path through the network and every single activity on the path must finish on time in order for the project to come in on time. Therefore the float for every activity in the critical path is always 0.

(ii) = 20

The total duration of path A-B-C-G is $6 + 4 + 8 + 2 = 20$

(iii) = 18

The total duration of path A-B-F-G is $6 + 4 + 6 + 2 = 18$

(iv) = 19

The total duration of path D-E-F-G is $7 + 4 + 6 + 2 = 19$

(v) = 17

The total duration for path D-H-I is $7 + 8 + 2 = 17$

Additional Precedence Diagram Samples

(vi) = A-B-C-G

A critical path is a string of activities that is longer than any other path through the network and every single activity on the path must finish on time in order for the project to come in on time. A-B-C-G is the longest path with the duration of 20.

(vii): A = 0, B = 0, C = 0, D = 1, E = 1, F = 1, G = 0, H = 3, I = 3

Evaluation of Learning Outcomes

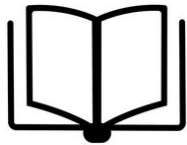
- Can you construct and analyse a network diagram from a WBS?
- Can you analyse and evaluate the early start and finish dates using the forward pass, and the late start and finish dates using the backward pass?
- Can you identify the critical path in a network diagram and carry out calculations of the free and total floats from a network diagram?
- Can you apply all of these to the semester project?

More on Precedence Diagramming Model (PDM):



YouTube

<https://youtu.be/-TDh-5n90vk>



Open source eBook: Watt, A. (2020). '10. Project Schedule Planning'. *Project Management*. Open Education. <https://opentextbc.ca/projectmanagement/chapter/chapter-10-project-schedule-planning-project-management/>

Project Management – A Systems Approach to Planning, Scheduling, and Controlling. 13th ed. John Wiley & Sons. [\(12: Network Scheduling Techniques\)](#)

The End

Next lecture

- Unit 10: Cost and revenue projections



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